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Modelling the non-linear response of Spanish electricity demand to temperature variations

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Abstract

The demand for electricity is a key variable because its links to economic activity and development; however, the electricity consumption also depends on other non-economic variables, notably the weather. The aim of this study is to analyse the effect of temperatures on the variability of the Spanish daily electricity demand, and especially to characterise the non-linearity of the response of demand to variations in temperature. In this article, we explore the ability of Smooth Transition (STR), Threshold Regression (TR), and Switching Regressions (SR) models, to handle both aspects. As we conclude, the use of LSTR approach offers two main advantages. First, it captures adequately the smooth response of electricity demand to temperature variations in intermediate ranges of temperatures. Second, it provides a method to analyse the validity of temperature thresholds used to build the "cooling degree days" (CDD) and "heating degree days" (HDD) variables traditionally employed in the literature.
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1. Introduction

Energy requirements, and specifically a country's demand for electricity, are strongly linked to climatological factors, which explain in large measure their intra-annual

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variability. Temperature stands out among the meteorological factors that affect electricity demand, as reported by Engle et al. (1986), Filippini (1995) Henley and Peirson (1997, 1998), Considine (2000), Johnsen (2001), Valor et al. (2001) or Pardo et al. (2002), among others. However, temperature is not the only variable considered in the literature. Thus Sailor and Muñoz (1997) or Yan (1998) use temperature data combined with data on humidity, wind speed, cloudiness, rainfall, and solar radiation to draw up a single climate variable that these authors relate to the demand for electricity.

The principal studies indicate that the relationship between electricity demand and temperature is clearly non-linear. This non-linearity refers to the fact that both increases and decreases of temperature, linked to the passing of certain "threshold" temperatures, increase the demand for electricity. According to Henley and Peirson (1997), this response is caused by the differences between the ambient or outdoor temperature and the comfort or indoor temperature. When the differential between outdoor and indoor temperatures increases, the starting-up of the corresponding heating or cooling equipment immediately raises the demand for energy. Naturally, the curve of the response of demand to temperatures depends especially on the climate characteristics of the geographical area to which the demand data refer, which condition the type of equipment installed, and the conditions under which it will start to operate. Thus, in especially hot areas, there will be hardly any response to cold, and conversely in relatively cold or temperate regions, there will be little or no response to heat.[1] Also, the notion of indoor temperature is not a parameter that can be considered stable or immutable, since variables such as income, consumer preferences or energy prices, can cause variations in it.

In dealing with the non-linearity of the temperature effect, the most frequent posture in the literature is to segment temperature variations in terms of the functions HDD or "*heating degree days*" and CDD or "*cooling degree days*". The HDD and CDD functions are defined conventionally in monthly terms as $\mathrm{HDD} = \sum_{j=1}^{\mathrm{nd}} \max\left(0; T^* - t_j\right)$ and $\mathrm{CDD} = \sum_{j=1}^{nd} \max\left(0; t_j - T^*\right)$, where nd is the number of days of a particular month, T^* is the threshold temperature[2] of cold or heat, and t_j the observed temperature on day j.

These functions reflect in combination the number of days on which the temperature stands below or above the thresholds of cold and heat, and by how many degrees. Since there is no strict quantification of the values of the "threshold" temperatures, there can be many different versions of the functions HDD and CDD. For example, Sailor and Muñoz (1997) consider a threshold temperature of 18.3 °C for both winter and summer, but the State of Florida, for which they consider a threshold temperature of 21 °C.

The objective of this study is to present a new approach for analysing the effect of temperatures on the variability of the Spanish daily demand for electrical energy on the Spanish, attempting to determine the sensitivity of demand to variations in temperature (Fig. 1). This approach involves the use of "smooth transition regression" (STR) models, widely diffused and used in the econometric literature on the business cycle, and whose

[1] On occasions the equipment predominating in the demand is for uses other than heating or cooling (e.g. de-humidifying apparatus) which may reduce the utility of temperatures as climate variables. In these, cases it is more appropriate to use a temperature–humidity combined index, such as HUMIDEX.

[2] T^* can be different for HDD and CDD, allowing for a temperature interval within which electricity demand is irresponsive to temperature variations.

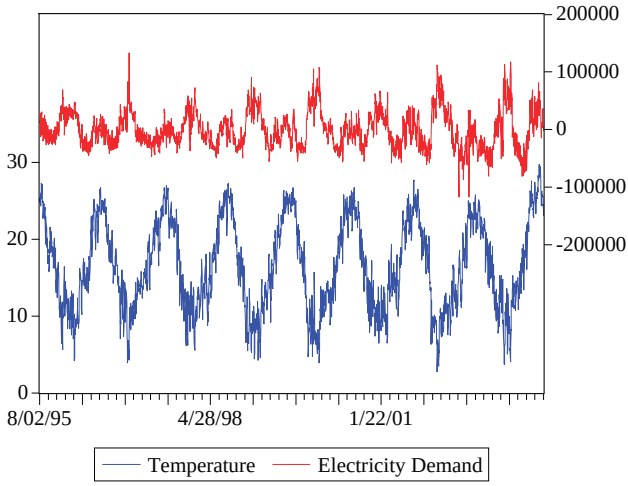

Fig. 1. Temperature (bottom graph) and filtered electricity demand (upper graph).

potentialities for the analysis of the response of electricity demand have not yet been exploited; this article being the first occasion, as far as we know, on which these models have been used for this purpose.

The next section describes in detail the variables used in this study, and the treatment of the electricity demand variable prior to the analysis of the effect of temperature. This analysis is presented in the third section, which analyses the main features of the response of electricity demand to variations in temperature by means of several non-linear methods, like "switching regression" (SR) or "threshold regression" (TR) models, and finally offers the results of estimating the proposed STR model. The article closes with the main conclusions derived from the study.

2. Data used

2.1. Electricity demand

The variable analysed is the aggregate Spanish demand for electricity (MWh) in with daily data, denoted by $D_{\mathrm{dd/mm/aa}}$ (demand on day dd of month mm of year aa), or in the abbreviated form D_t. The period studied is that between August 1995 and August 2003. Electricity demand exhibits a persistent trend and seasonality; also, given the daily frequency of the data analysed, the predominance of the working day effect is patent. This aspect was previously highlighted by Pardo et al. (2002) and Valor et al. (2001), among others.

The calendar effect is collected by a variable called "working day effect", WD_t, which represents (independently of the rest of effects as seasonal, temperature effect, trend and economic conditions) the effect of calendar in the demand of a particular day as a percentage of electricity demand on a representative day. In this study, we use Wednesday as the representative day of the week, assigning it a calendar factor equal to one. The method we

use to obtain the calendar factor consists the following steps. First, we estimate the weekly calendar variation index, WCVI, as WCVI=$D_{d/w}/W_w$, where $D_{d/w}$ is the electricity demand on day d of week w, and W_w is the electricity demand on Wednesday[3] of the same week w. Since the effects of trend, seasonality, temperature and economic conditions are smooth, and $D_{d/w}$ and W_w are sufficiently near, we can conclude that all the effects cancel out, except for the calendar effect which is captured by the WCVI variable. The second step is the determination of homogeneous types of day. This step basically consists in regressing the WCVI variable on a combination of dummies reflecting the week of the day (Monday, Tuesday,. . ..) and the types of holidays,[4] and then, to group the holidays and/or week days with similar values of the WCVI.[5]

As a result of these steps, we obtain the new variable "working day effect", which captures the electricity demand variation caused by the different level of activity in the industrial and commercial sectors and in the behaviour of households on holidays and on working days. The values of the variable "Working day effect", assigned to each type of day are collected in Table 1, these values represent the coefficient of equivalence of the different types of day, which would "transform" any type of day into a "typical"[6] or most usual day. With this variable, we can eliminate from the demand series the effects of the calendar, which would "mask" those of the rest of the components of demand.

The results presented in Table 1 are congruent with those obtained by Pardo et al. (2002), despite the differences in the methodology employed and in the types of days considered.[7] Both studies show that the days of the week with less demand are, in decreasing order, Monday, Saturday and Sunday. Pardo et al. (2002) find that the days with high demand: Tuesday, Wednesday and Thursday, also exhibit a similar electricity demand level. Similarly, they show that holidays are associated with large declines in the demand. This behaviour can also be seen in the coefficients included in Table 1.

2.2. Temperature

Although it is more appropriate to carry out an analysis of electricity demand disaggregated by sectors and geographically, this is unviable given the limitations of the data used in this study. The lack of a suitably disaggregated demand series means that the

[3] If W_w is a holiday, we take the mean of the electricity demand on the previous and posterior (working day) Wednesdays.

[4] Typologies considered are: non-holidays, national holidays, local holidays, days following a national holiday, days following a local holiday, Easter and Christmas).

[5] In this work, we have employed the Wald test to test if the differences in the estimated value of the coefficients assigned to each dummy are negligibles. In such case, we consider that the two holidays and/or the two types of week days are similar. As an example, consider the case of non-holiday type of day. We select all the values of the WCVI variable that correspond to days that are not holidays, then we perform a regression between WCVI and seven dummies, each for every day of the week, without intercept. After that, we use the Wald test to determine if, for example, the value of the estimated coefficient for Tuesdays is, say, equal to those estimated for the Wednesdays.

[6] As previously pointed out, the type-day is a working Wednesday, which acts as benchmark for classifying all the possible days that might occur during a year.

[7] See Pardo et al. (2002) for further details on the methodology and the holidays considered in their study.

Table1
Working day effect

Type of day	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Non-holidays	0.97	1.00	1.00	1.00	0.99	0.88	0.78
Easter							
Week before Easter day	0.96	0.98	0.96	0.83	0.73	0.77	0.72
Easter Monday and Easter Tuesday	0.83	0.96	–	–	–	–	–
Christmas							
21th–22th December and 3rd–4th January	0.93		0.97			0.85	0.75
23th, 27th, 28th, 29th, 30th December	0.85		0.88			0.80	0.75
25th December				0.66			
1st January				0.67			
6th January				0.76			
26th December, 2nd and 7th January			0.86			0.79	0.73
24th December			0.78			0.71	
31th December			0.75			0.71	
5th January	0.78		0.93			0.82	
National holidays							
1st May		0.72		0.76		0.79	0.72
15th August	0.80	0.85					0.82
12th October	0.76		0.82			0.83	0.76
1st November	0.77		0.80			0.86	0.77
6th December	0.79		0.87			0.89	0.79
Day after national holidays							
16th August	0.92	0.98		0.92		0.88	0.81
13th October	0.96				0.90	0.79	
2nd November	0.95				0.87	0.83	
7th December	0.83	0.88	0.96	0.91		0.88	0.83
Monday before national holidays	0.88	–	–	–	–	–	–
Regional and local holidays							
25th July, 24th June, 11th September	0.90	0.95				0.84	0.76
28th February, 23th April, 9th October, 15th May, 9th November, 24th September	0.95	0.97	0.98		0.97	0.85	0.79
19th March, 2nd May	0.90		0.94	0.89	0.84	0.80	0.76
8th December	0.79		0.82				
Day after regional and local holidays							
26th July, 25th June, 12th September	0.99	0.99	1.00		0.96	0.84	0.76
29th February/1st March, 24th April, 10th October, 16th May, 10th November, 25th September	0.96		1.00			0.87	0.75
20th March, 3rd May	0.94	0.99	0.97	0.99	0.94	0.83	0.76
9th December	0.96		0.99		0.90	0.83	0.81

Coefficients rounded to 2 decimal places.

explanatory variables have to be considered in the aggregate; likewise the behaviour equations and the parameters obtained reflect aggregated or average responses.

In the study of the temperature sensitivity of electricity consumption, these limitations of information give rise to two essential points that may inspire criticism. On the one hand, the aggregated study requires an aggregate explanatory variable, i.e. a mainland-wide indicator of temperature in Spain, and on the other hand, we have to assume a homogeneous response to temperature variations in the different sectors (e.g. residential, industry and services).

Spain does not have a homogeneous climate, so it is not possible to determine immediately a single indicator of temperature. On the basis of this limitation, the procedure for obtaining a representative temperature indicator is as follows:

1. Determination of homogeneous climate zones. On the basis of the mean daily temperature values[8] registered in at the 47 observatories in the provincial capitals, 7 homogeneous geographical zones were identified by means of a cluster analysis[9] (see Table 2).
2. Selection of representative observatories of each climate zone, according to its distance from the centre of each homogeneous group.
3. Calculation of the mainland-wide temperature indicator. Once the observatories have been selected, an aggregate mainland-wide temperature measurement is obtained, as a weighted mean of the temperatures registered in each representative observatory. The weightings used in this case are obtained from the population figures according to the 2001 population census of Spain.[10]

2.3. Treatment of trend and seasonality

Since the aim of this study is to analyse the response of demand to temperature variations, the existence in the demand of a trend and seasonality not linked to climate conditions[11] creates the necessity of first treating these components.

Even though other alternatives can also be considered suitable,[12] we consider that the trend can be represented adequately by means of a third degree time polynomial, and that the non-climatic seasonality can be represented simply by a dummy in the month of August. These elements, trend and non-climate seasonality, together with the working day effect, make up what we will call the deterministic component of the demand, which will be "eliminated" from the demand series prior to the analysis of sensitivity to temperature.

[8] Simple mean of the maximum and minimum temperature values registered on a particular day at a particular observatory.

[9] Cluster analysis is performed using Ward's Method (Euclidean distance). We first identify eight groups, then we recombined them in seven groups by using geographical proximity criteria and validate them with an ANOVA test. The representative observatories are the nearest observatories to centroids of such groups, using weighted population distances.

[10] This is the more recent official census (the previous census dates back to 1991).

[11] We refer specifically to the seasonality caused by the summer holidays, which generates a considerable reduction in electricity demand in the month of August, mainly in the industry activities.

[12] See, for example, Hamilton (1994).

Table 2
Climatic zones and temperature statistics in each representative observatory

Climatic zones	Provinces included in each zone	Zone weight	Representative observatory	Observatory: temperature statistics 1995/8–2003/8			
				Mean	S.D.	Max.[a]	Min.[b]
Z 1	Albacete, Ciudad Real, Lérida, Toledo, Granada, Zaragoza, Madrid	0.232	Madrid	14.9	7.2	30.7	−2.4
Z 2	Alicante, Castellón, Valencia, Murcia, Almería	0.154	Valencia	18.5	5.7	31.4	3.7
Z 3	Avila, Burgos, León, Palencia, Soria, Alava, Cuenca, Guadalajara, Huesca, Logroño, Navarra, Salamanca, Segovia, Teruel, Valladolid, Zamora	0.112	Valladolid	13.4	6.9	31.0	−4.8
Z 4	Badajoz, Jaén, Cáceres, Sevilla, Córdoba	0.110	Sevilla	19.9	6.2	36.2	5.9
Z 5	Lugo, Coruña, Orense, Pontevedra, Santander, Guipúzcoa, Oviedo, Vizcaya	0.160	Bilbao	15.2	5.0	34.3	−0.3
Z 6	Gerona, Tarragona, Barcelona	0.157	Barcelona	16.7	5.7	30.9	1.6
Z 7	Cádiz, Huelva, Málaga	0.075	Malaga	19.1	5.3	33.4	7.6

[a] Maximum of the daily mean temperature registered in the observatory.
[b] Minimum of the daily mean temperature registered in the observatory.

To filter these components, we estimate by OLS ("ordinary least squares") the model:

$$D_t = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3 + \alpha_4 I_{\text{august}} + \alpha_5 \text{WD}_t + \text{DF}_t \quad (1)$$

where D_t is the aggregate demand for electricity, t is a time variable (t=0, 1, 2,. . .), I_{august} is a dummy variable taking the value 1 if the observation of the demand corresponds to the

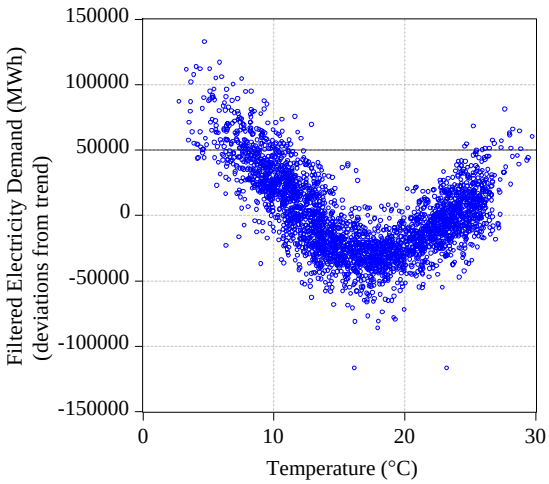

Fig. 2. Non-linearity in electricity demand response to temperature variations.

month of August, WD_t is the "working day effect" variable previously detailed and DF_t is the residual component, or electricity demand with the deterministic component filtered out.

With the deterministic components filtered out, it is possible to appreciate, by means of a simple graphic analysis, the non-linearity of the response of electricity demand to variations in temperature. As shown in Fig. 2, the response curve[13] presents an asymmetrical V shape with a minimum around 18 °C, a shape similar to that already noted in the abovementioned studies by Engle et al. (1986), Sailor and Muñoz (1997), Valor et al. (2001) or Pardo et al. (2002).

3. Analysis of the electricity demand response to temperature with non-linear models

The non-linearity of the response of electricity demand to temperatures is caused by the fact that demand increases when the temperature decreases (with "low" temperatures) and when it increases (with "high" temperatures). In order to incorporate this behaviour into a linear model, the traditional approach has to been to segment the variation of temperatures into two new variables, therefore segregating the behaviour of demand when temperatures are "cold" and when temperatures are "hot".

On the basis of this segmentation, we construct the auxiliary variables HDD and CDD, which capture the energy requirements for "heating" ("heating degree days") when temperatures are low and the requirements for "cooling" ("cooling degree days") when temperatures are high. Both variables enable us to construct a linear model that avoids the limitations deriving from the behaviour of the demand.

Despite the good results produced by this method, it is not without its limitations. In particular, this procedure requires us to identify "cold" and "hot" temperatures on the basis of passing a certain temperature threshold, or certain temperature thresholds, usually quantified by simple inspection of a graph, and whose validity is not subjected to any kind of check. Nor is it a trivial matter whether a single threshold or a dual threshold is/are considered, since in the first case, it would mean that when the threshold was passed in one direction or the other, there would automatically be a sharp change in the behaviour of the demand, whereas considering two thresholds allows us to assume that for the intermediate temperature range there is no appreciable change in demand.

Trying to overcome these limitations, in the following sections, we explore the ability of three models[14] to capture the non-linear features of the electricity response to temperatures, focusing on the way these models capture the relation between temperature and "switch" in the qualitative electricity demand behaviour. These models fall into the category of "alternating" or "switching" models according to which a time series can be characterised on the basis of different linear models that prevail in a certain state and follow each other in succession or alternate with each other. The characteristics of these models adapt adequately to the behaviour of electricity demand, in that the different states can be associated immediately with temperature conditions.

[13] Negative values emerge because the data plotted (DF_t) are obtained detrending the original series as explained in Section 2.3. Since DF_t is a "residual" component obtained as the resid of a regression estimated by ordinary least squares, it has zero mean, and therefore, some values can be negative.

[14] Switching Regression (SR), Threshold Regression (TR) and Smooth Transition Regression (STR) models.

3.1. Switching regression model

Since the response of demand to "cold" temperatures differs from that to "hot" temperatures (due to differences in use, type of equipment or the technology used), we will assume that the response of electricity demand[15] (DF_t) to temperature[16] (TMP_t) can be represented generically on the basis of a linear model with two states, given by the expression:

$$DF_t = \mu_{St} + TMP_t\beta_{St} + \varepsilon_t, \tag{2}$$

with $\varepsilon_t \sim N(0,\sigma^2_{St})$, and St=1,2 representing each of the two states ("cold" or "hot" temperatures) that the demand series may present, and β_{St} is the parameter that captures the effect of temperature on electricity demand when state is "cold" or "hot" (μ_{St} is a intercept term that usually do not have a meaningful interpretation).

The starting point is to assume that demand may be in either state of those between which it independently alternates, i.e. we assume random switch without imposing a priori any relation between switch and the temperature variable.

Assuming that the probability of being in a particular state is independent of past values, and denoting the information available up to period t as ψ_{t-1}, we will find that $PR(S_t=1|\psi_{t-1})=PR(S_t=1)=p$ and therefore $PR(S_t=2)=1-p$. On this basis,[17] and assuming normality in the disturbance, $\varepsilon_t \sim N(0,\sigma^2_{St})$, the logarithm of the likelihood function of the data would be given, with T observations, by:

$$\ln L = \sum_{t=1}^{T} \ln\left[\sum_{St=1}^{2} f(DF_t|S_t,\psi_{t-1})\Pr(S_t|\psi_{t-1})\right] \tag{3}$$

where $f(DF_t|S_t,\psi_{t-1}) = \left(1/\left(\sigma_{St}\sqrt{2\pi}\right)\right)\exp\left[-\frac{1}{2}(DF_t - \mu_{St} - TMP_t\beta_{St}/2\sigma_{St})^2\right]$ and $PR(S_t|\psi_{t-1})$ is the probability that the model is in state S_t (1 or 2) given the information available up to period t (ψ_{t-1}).

This model, can be easily estimated by means of the expectation–maximization EM algorithm detailed in Quandt (1988).[18] The switching regression (SR) model gives a first approximation to the determinants of the "change" of state and their relation with the temperature variable; for this purpose, we estimate the model by means of the EM algorithm, "recovering" the probabilities of state from step E ("*expectation*") of the algorithm, which are given by the expression:

$$\Pr(S_t = i|\psi_t) = \frac{f[DF_t|S_t = i]\Pr[S_t = i]}{\sum_{St=1}^{2} f(DF_t|S_t)\Pr(S_t)} \quad (i = 1,2) \tag{4}$$

Expression (2) provides the Bayesian probability that the observation of the variable DF_t has been generated by the model prevailing in the ith state.

Estimating the model and calculating expression (4), we depict these probabilities in relation with the temperature in Fig. 3. As can be appreciated from this graph, without

[15] Electricity demand filtered for deterministic effects.

[16] A vector of climate, economic or other type of variables can be considered.

[17] Note that the model does not impose any restriction over the determinants of changes in state. In this model, the temperature is used to explain the demand but not to explain why the model switches from one state to another.

[18] The algorithm calculations are detailed in Appendix A.

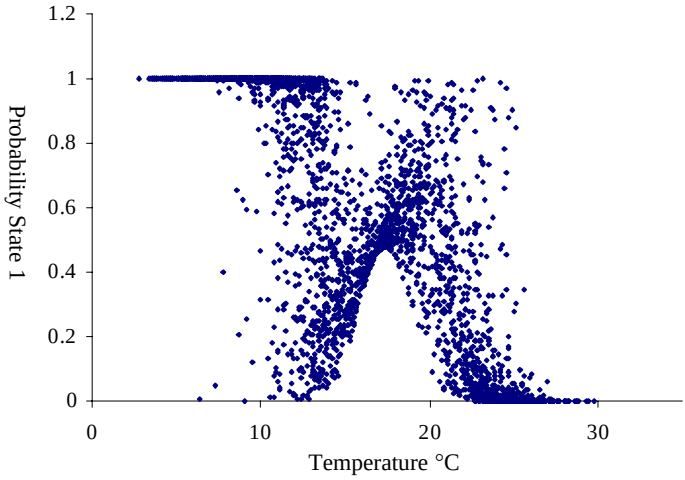

Fig. 3. Probability state 1 vs. temperature.

imposing restrictions on the model, the link between temperature and "state" is manifest. One of the two states presents a high probability of occurrence when temperatures are below 13 °C,[19] and a very low probability when the temperature exceeds 20 °C, permitting us to identify this state with that corresponding to "cold" temperatures. Between these temperatures, there exists an "indetermination" area, in which the probabilities of being in a particular state are in the interval 0.4–0.6, a situation that corresponds to the seasons of spring and autumn.

As well as the above characteristics, we can observe a zone of smooth transition from one state to the other, in which the probability of state varies progressively without sudden jumps. This behaviour is congruent with the progressive substitution of equipment specific to "winter" or "cold" by that specific to "summer" or "heat" as temperatures increase, a process that is gradual though with frequent "reverses" in the passage from one state to the other.

3.2. Two threshold regression (TR) model

A two threshold model is conceptually similar to a linear model with three regimes or states, in which the transition from one to another occurs as the threshold variable (temperature, in this case, TMP_t) takes the values defined by the thresholds Th1 and Th2 (Th2>Th1). The model is:

$$\mathrm{DF}_t = \begin{cases} \mu_1 + \beta \mathrm{TMP}_t + \varepsilon_t & \mathrm{TMP}_t < \mathrm{Th1} \\ \mu_2 + \gamma \mathrm{TMP}_t + \varepsilon_t & \mathrm{Th2} > \mathrm{TMP}_t > = \mathrm{Th1} \\ \mu_3 + \eta \mathrm{TMP}_t + \varepsilon_t & \mathrm{TMP}_t > = \mathrm{Th2} \end{cases} \tag{5}$$

with $\varepsilon_t \sim i.i.d.N(0,\sigma)$. The estimation of these models is usually done (Tsay, 1989, 1998, Hansen, 2000) using consecutive estimations by ordinary least squares (OLS), modifying at

[19] Since the states are complementary, a high probability associated with one state necessarily implies a low probability of the other state.

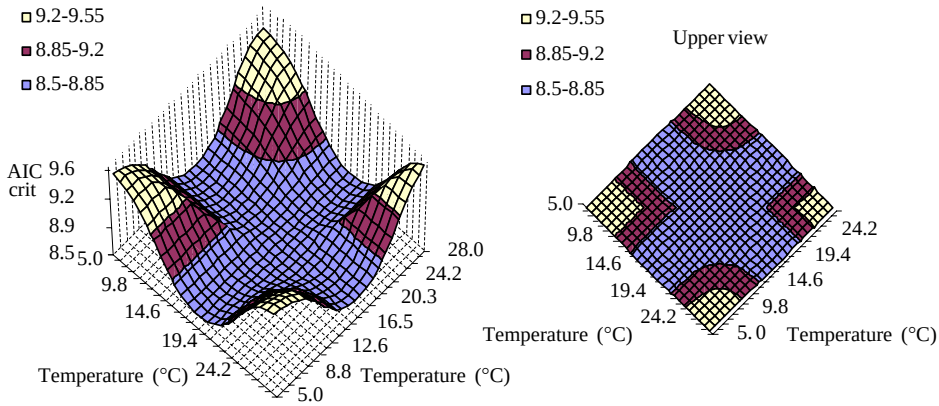

Fig. 4. AIC and temperature thresholds.

each step the value of the thresholds, obtaining the "Akaike information criterion" (AIC) of the estimation or, alternatively, the sum of squared errors. The AIC is used to determine the optimal threshold values, being those that provide the lowest AIC value. Fig. 4 represents the AIC in terms of different values of temperature threshold 1 (Th1) and of threshold 2 (Th2).

The gradual transition between the "cold" and "heat" regimes previously detected with the switching regression (SR) model is corroborated when we analyse the value of the AIC in the two threshold model of electricity demand against temperature, taking the temperature itself as the threshold variable. As shown in Fig. 4, there exists a "broad" region of upper and lower threshold temperature values that offer very similar AIC values for the dual threshold model analysed, which is indicative of a smooth transition from one state to another, since if there were a sudden transition between states, the values of the thresholds that minimise the AIC would be more "defined". As an interesting result, with the data used in this study, the minimum AIC[20] is obtained for a lower threshold of 15.5 °C and an upper threshold of 18.4 °C.

3.3. STR model of the response of electricity demand to temperature

The results of the analysis presented in the above section suggest that the transition between the equipment for "cold" and that for "heat" is gradual and does not occur suddenly. This behaviour is "captured" adequately by the so-called "smooth transition" or STR models (see Teräsvirta and Anderson, 1992; Granger and Teräsvirta, 1993; Teräsvirta, 1994, 1998), in which the probability of prevalence of one state is given by a continuous generic function,

[20] With Th1=15.5 °C and Th2=18.4 °C the estimation results of TR model (5) are :

	μ_1	μ_2	μ_3	β	η
Parameter estimates (*)	123.2187	−30.39666	−154.9104	−9.764103	6.540661
	(2.123054)	(0.937661)	(5.199766)	(0.183893)	(0.227754)
S.E of regression ($\hat{\sigma}$)	18.44595		Akaike info criterion AIC	8.669259	
Log-likelihood	−12790.83		Schwarz criterion SC	8.679406	

(*) Std. errors of parameters in parenthesis. The model has been estimated using ordinary least squares method given Th1 and Th2. The parameter γ in expression (5) is set to zero.

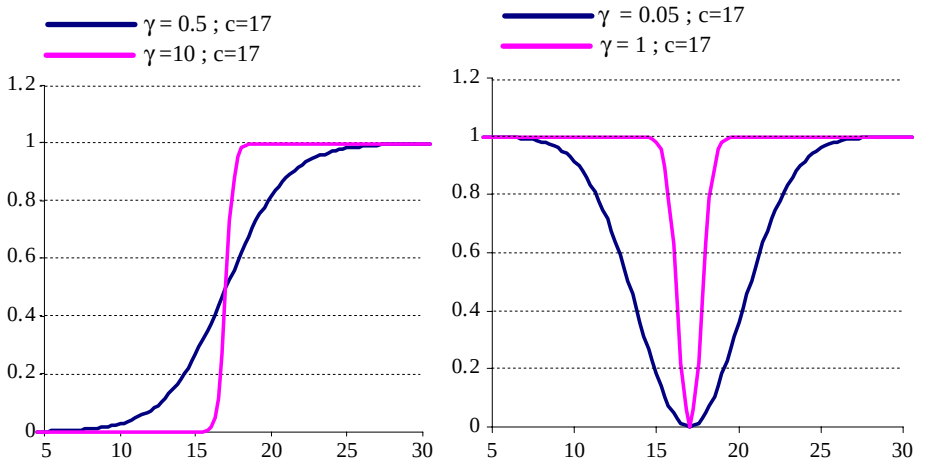

Fig. 5. Examples of Transition functions (analytic expression). Logistic (analytic expression) (left). Exponential (analytic expression) (right).

$G(z_t,\gamma,c)$, restricted to the interval [0,1], which is dependent on a transition variable, z_t, and on the parameters γ and c. The latter parameter determines for which values of the variable z_t, the change of state occurs and on occasions is called the "threshold". The parameter γ measures the smoothness of transition between states, as γ becomes very large the transition is more abrupt, so the model could be substituted by a TR model with just one threshold.

In the presented study, we opt for a model of the "logistic smooth transition" type (LSTR), because the logistic function adequately represents the passage from one state in the demand for electricity (corresponding to "cold temperatures") to the other ("hot temperatures"), with an area of uncertainty in between where neither predominates. "Exponential smooth transition" models (ESTR) were ruled out because with an exponential function (see Fig. 5 for the comparison between the two types of function), the two states would be associated, respectively, with a situation of extreme temperatures and with one of "average" temperatures; it cannot, however, be accepted a priori that the electricity demand corresponding to cold temperatures and to hot temperatures could fall within the framework of a single "state" or regime.

The LSTR model proposed for modelling the response of electricity demand to temperatures can be expressed (Teräsvirta, 1994) according to:

$$\mathrm{DF}_t=(\alpha_1+\beta_1\mathrm{TMP}_t)[1-G(\mathrm{TMP}_t;\gamma,c)]+(\alpha_2+\beta_2\mathrm{TMP}_t)G(\mathrm{TMP}_t;\gamma,c)+v_t \qquad (6)$$

where in Eq. (6), it has been assumed that the transition variable is the average temperature registered on the same day, TMP_t, the transition function being of the logistic type $G(\mathrm{TMP}_t;\gamma,c)=[1+\exp\{-\gamma(\mathrm{TMP}_t-c)\}]^{-1}$. For simplicity, we will suppose that v_t is distributed identically and independently as a normal random variable[21] with zero mean and variance given by σ^2.

[21] The normality of the disturbance is not an essential requirement. This restriction is imposed simply to estimate the model by maximum likelihood assuming a normal distribution.

Table 3
Estimation results LSTR[a] model $DF_t=(\alpha_1+\beta_1 TMP_t)[1-G(TMP_t;\ \gamma,c)]+(\alpha_2+\beta_2 TMP_t)G(TMP_t;\gamma,c)+v_t$; $G(TMP_t;\gamma,c)=[1+\exp\{-\gamma(TMP_t-c)\}]^{-1}$

Parameter	Estimation method	
	Maximum likelihood	Non-linear least squares
α_1	105.4998 (5.312440)	105.5003 (6.170249)
α_2	−6.479369 (1.136533)	−6.479481 (1.227945)
β_1	−269.9439 (10.54021)	−269.9432 (10.44514)
β_2	13.47021 (0.774074)	13.47027 (0.829253)
γ	0.404822 (0.037710)	0.404824 (0.037162)
C	15.41746 (0.650561)	15.41751 (0.686990)
S.E. of regression ($\hat{\sigma}$)	18.37945	18.39504
Log-likelihood	−12,782.17	−12,782.17
Akaike info criterion AIC	8.665	8.664
Schwarz criterion SC	8.679	8.676

[a] Std. errors of parameters in parenthesis.

The estimated values of the parameters and their standard deviation are shown in Table 3; the estimation used the non-linear least squares (NLS) method, proposed by Teräsvirta (1994, 1998), and the maximum likelihood (ML) method.

According to the LSTR model estimated, the response of electricity demand presents a change of behaviour from 15.42 °C, the temperature at which the transition function reaches a inflexion point, or alternatively, the "probability" that the observed response in demand corresponds to one of the states is equal to 0.5. The model estimated allows us to obtain a "theoretical" representation of the electricity consumption at each temperature,

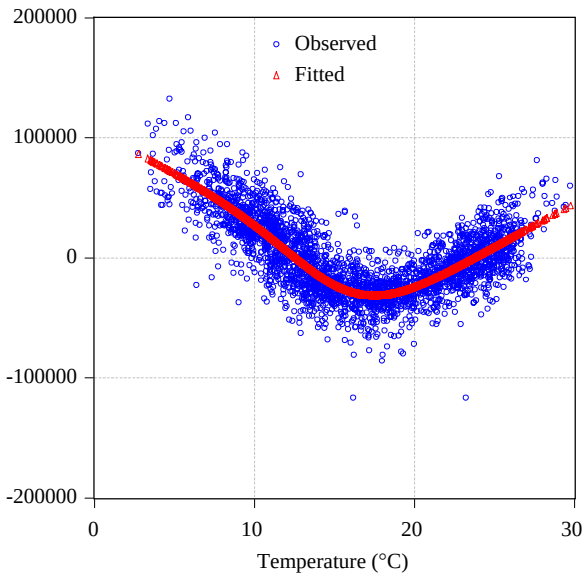

Fig. 6. Electricity response observed and fitted.

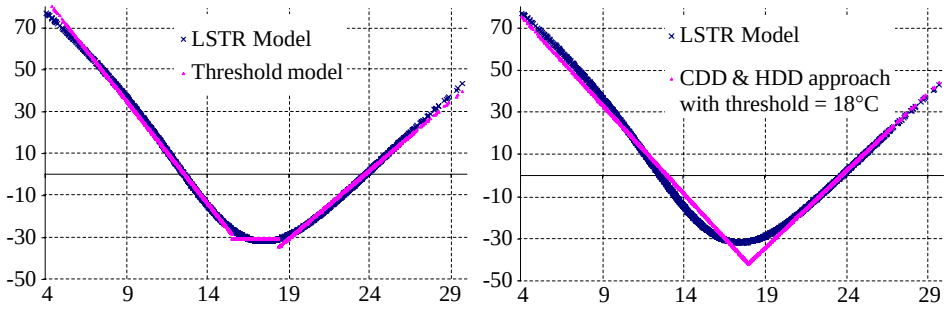

Fig. 7. Electricity response to temperature variation and estimated LSTR, TR models and CDD-HDD based modelling approach. In the two-threshold regression model (TR), the temperature thresholds are 15.5 °C and 18.4 °C.

which is reflected in Fig. 6, which also represents the observed response of electricity demand to variations in temperature.

The LSTR model adequately captures the non-linearity present in the demand for electricity, as well as the smooth transition occurring from a regime characterised by cold temperatures to a regime corresponding to hot temperatures. This characteristic is the main difference between the model presented here and an approach based on the threshold regression model (TR) or the switching regression model (SR) previously analysed, though the theoretical response of both approaches[22] is fairly similar, as shown in Fig. 7.[23] The parameters γ and c in LSTR models parsimoniously define the observed response of electricity demand to temperature variations with flexibility. However, in SR model, as is showed in Fig. 3, the relation between temperature and "state" transition was more "diffuse", given by the smoothness exhibited by the demand response. In the same way, in TR models the reliability of the threshold estimates would be affected by the smoothness of demand response. If the demand response is "flat", the range of threshold values that work similarly well, increases. So, the LSTR model can serve additionally to test the reliability of the threshold employed to build the "cooling degree days" (CDD) and "heating degree days" (HDD) variables traditionally employed in the literature.

Another potential use of this approach is to analyse the temporal or spatial[24] stability of the electricity demand response to temperatures, comparing the estimated values of the parameters in different dates or regions. This analysis can be used, for example, in demand

[22] Considering a single threshold model would mean using a "V" shaped theoretical response curve, whereas a two threshold model is associated with a response curve similar to a truncated "V", as shown in Fig. 7.

[23] In Fig. 7 is also included the estimated response using the CDD and HDD based modelling approach. In this case, the model estimated is: $DF_t = \alpha_1 + \beta_1 CDD_t + \beta_2 HDD + \xi t$, with a threshold temperature of 18 °C as is derived from the visual inspection of Fig. 2. The estimation results are:

	α_1	β_1	β_2
Parameter estimates (*)	−41.76093	8.384833	7.318080
	(0.700117)	(0.114925)	(0.158120)
S.E of regression ($\hat{\sigma}$)	18.73571	Akaike info criterion AIC	8.699755
Log-likelihood	−12837.84	Schwarz+criterion SC	8.705843

(*) Std. errors of parameters in parenthesis. The model has been estimated using ordinary least squares method.

[24] Regional or country comparisons.

Table 4
Estimation results LSTR[a] model with a three-years rolling sample $DF_t=(\alpha_1+\beta_1 TMP_t)[1-G(TMP_t;\gamma,c)]+(\alpha_2+\beta_2 TMP_t)G(TMP_t;\gamma,c)+v_t$; $G(TMP_t;\gamma,c)=[1+\exp\{-\gamma(TMP_t-c)\}]^{-1}$

Parameter	Sample period											
	8/02/1995–8/31/1998		8/02/1996–8/31/1999		8/02/1997–8/31/2000		8/02/1998–8/31/2001		8/02/1999–8/31/2002		8/02/2000–8/31/2003	
α_1	79.01056	(7.852892)	90.45193	(7.636317)	116.1098	(4.886419)	126.2311	(4.312408)	110.3544	(10.87336)	116.1891	(10.57415)
α_2	−4.967402	(1.095782)	−5.953708	(1.200276)	−8.795614	(0.616083)	−10.12202	(0.454695)	−6.118343	(2.306540)	−6.820370	(2.231822)
β_1	−187.4678	(7.960024)	−204.2049	(7.982032)	−214.8064	(17.63893)	−132.9423	(60.09580)	−272.8798	(15.12849)	−332.8462	(17.32817)
β_2	9.451662	(0.899789)	10.65315	(0.898726)	12.96503	(0.371220)	10.88444	(2.043747)	12.88536	(1.793563)	16.10195	(1.643446)
γ	0.580252	(0.097291)	0.518370	(0.082258)	0.555921	(0.084498)	0.525416	(0.073781)	0.435269	(0.064166)	0.400976	(0.051940)
c	15.31678	(0.609606)	15.36022	(0.776139)	17.02692	(0.965968)	18.81267	(0.886097)	14.17164	(0.875285)	15.07376	(0.959369)
S.E. of regression ($\hat{\sigma}$)	14.37287		14.58493		15.40713		16.78308		19.11800		21.48843	
Log-likelihood	−4595.892		−4608.285		−4674.136		−4770.455		−4917.126		−5044.252	
Akaike info criterion AIC	8.173876		8.203174		8.312852		8.483934		8.744452		8.978225	
Schwarz criterion SC	8.200660		8.229976		8.339636		8.510717		8.771236		9.005028	

[a] Std. errors of parameters in parenthesis. The model has been estimated using non-linear least squares method.

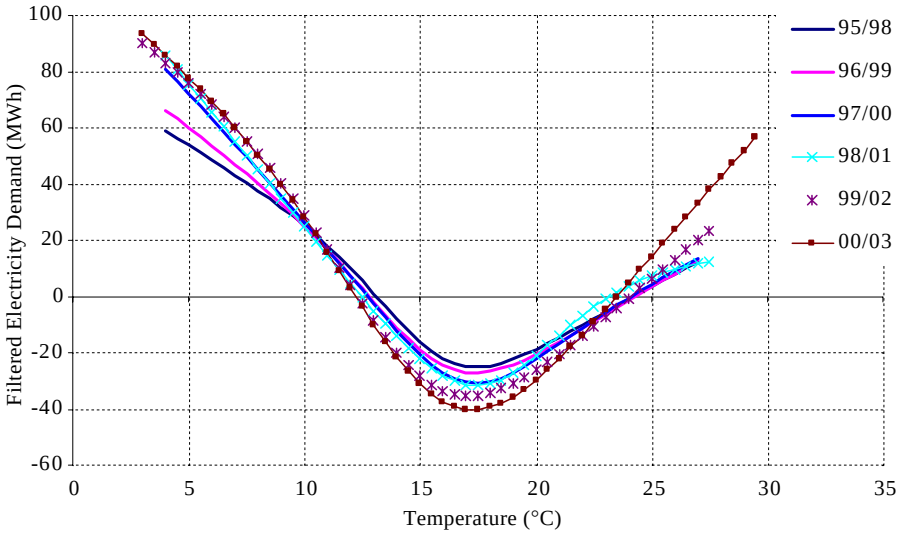

Fig. 8. Estimated demand response to temperatures by sample period. LSTR Model.

forecasting models, since the changes in the sensitivity of electricity demand to temperatures throughout the time could make the forecasting model more inestable, giving rise to systematic prediction errors.

In Table 4, we present the estimated results of the LSTR model using a 3-year rolling sample, graphing the estimated demand responses in Fig. 8. As we can see, the estimated response shows slight variations throughout the time, mainly concentrated in the slope of the demand response in each state (β_1 and β_2 parameters). This result indicates that increases in electricity consumption with high and low temperatures are becoming more pronounced, suggesting that the use and diffusion of heating and cooling equipment installed in Spain are more intensive in recent years.[25]

In any case, the estimated demand response may also be significantly affected by others factors not included explicitly in the model, like economic activity, variations in electricity relative prices, technology installed, etc.[26]. On the other hand, it is possible that the temperature for extended periods, not only the temperature registered in only a day,[27] could help to explain these signaled findings. For example, long periods of temperatures above or under normal ranges would reinforce the equipment accumulation dynamics, or more simply, would act like an incentive for the acquisition of support equipment (more

[25] In summer 2003, Spain was affected by a heat wave of unusual duration. The sales of air-conditioned equipment were boosted by this "natural" phenomenon as the AFEC data reflect (Asociacion de Fabricantes de Equipos de Climatización, http://www.afec.es).

[26] As example, in the Short-Term Energy Outlook of the U.S. Energy Administration Information "the main determinants of electricity demand are: household growth (residential sector); changes in commercial employment (commercial); growth in manufacturing output (industrial) or in real GDP (other); weather (residential and commercial); general seasonal factors (all) and trends in demand intensity (residential, commercial and other)." (http://www.eia.doe.gov/emeu/steo/pub/document/textel.html).

[27] Pardo et al. (2002) find that the relationship between temperature and electricity load is affected by a dynamic component of temperature.

intensive in electricity consumption) that only would be used in exceptional circumstances, increasing the observed response of electricity demand to temperature.

4. Conclusions

The usual treatment of the non-linearity of the response of electricity demand to temperatures consists of segmenting temperature variations through the functions "*heating degree days*" (HDD) and "*cooling degree days*" (CDD). These functions capture the number of days on which the temperature is below or above the thresholds of cold and heat, and by how many degrees the temperature has deviated from the thresholds. The wide dissemination of these functions contrasts with the lack of a suitable procedure for determining the values of the "threshold" temperatures, and testing the validity of the use of the functions CDD and HDD. This article has explored the potentialities of different non-linear models both in the analysis of the behaviour of demand and in its modelling and validation of temperature thresholds. Among the models considered, this article has shown the capacity of a simple LSTR model to capture the main features of the response of demand to temperatures, particularly, the threshold temperature at which the qualitative behaviour of electricity demand changes, and the degree of smoothness of the demand response, aspects which are closely linked. To show the potentialities of this approach, we have estimated the LSTR model using several sub-samples to explore the temporal stability of the demand response to temperature variations.

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Appendix A. Annex

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.eneco.2005.01.003.

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